

The Economy

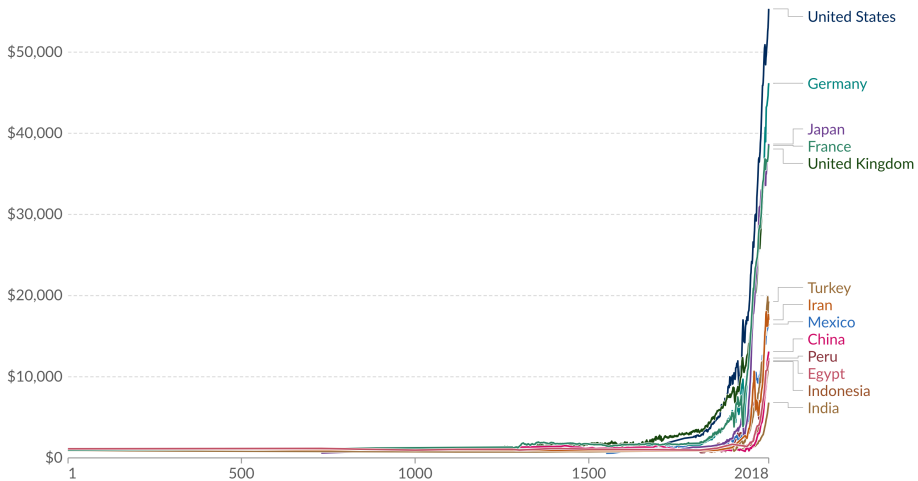
Seminar 2: Long-term Growth

Emir Tarık Dakın

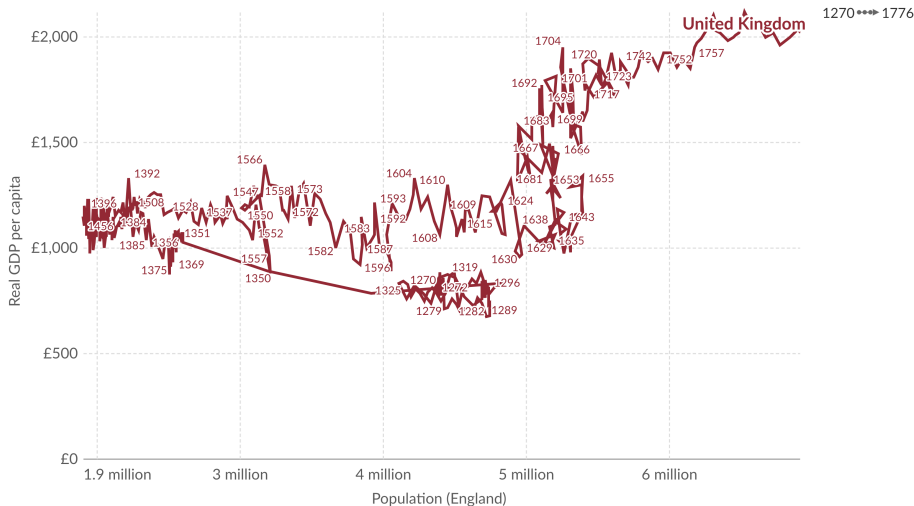
Sciences Po Paris - Reims

February 14, 2024

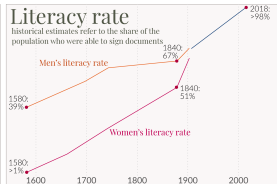
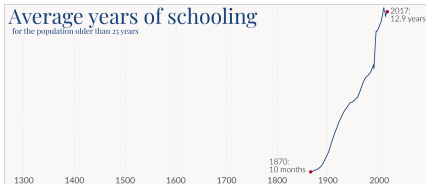
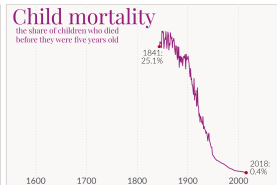
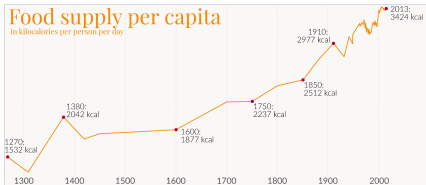
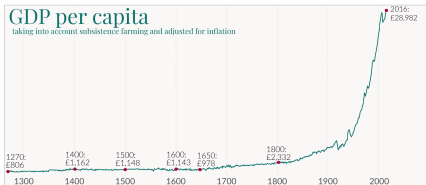
Malthusian trap



Malthusian trap



Malthusian trap



Production functions

The relationship between the quantity of **inputs** used in production and the quantity of **output** from production.

$$Y = AF(K, L)$$

- Y - Quantity of output.
- A - Technology. As technology improves, A rises, so the economy produces more output from any given combination of inputs.
- $F(\)$ - Function that shows how the inputs are combined to produce output.
- K - Quantity of physical capital.
- L - Quantity of labor.

- We measure with **increasing, constant and decreasing** returns to scale whether the output increases by a **larger, equal or smaller** amount when all the factors of productions are increased.
 - Malthusian, **decreasing returns to scale**: $F(mK, mL) < mF(K, L)$
 - Solow, **constant returns to scale**: $F(mK, mL) = mF(K, L)$

The Malthusian model: Assumptions

For this model: we set land to be constant and exogenous: $\overline{K}$, and have farmers as labor L as endogenous.

→ so the production function is simply $Y = AF(L)$

Key assumptions.

① decreasing returns to scale for labor:

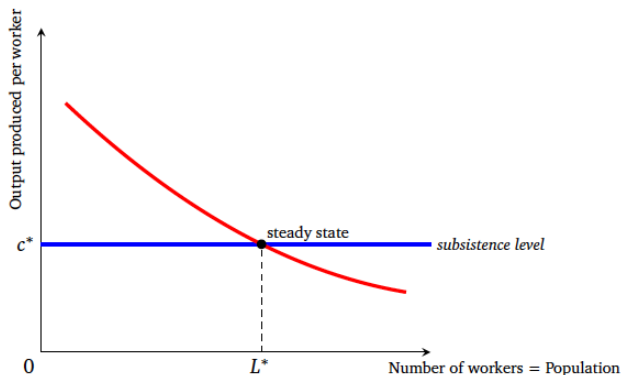
- as the number of farmers L , increases by 1 unit, the output per farmer $y = \frac{Y}{L}$ increases by less than 1 unit

② population expands if living standards (wages) increase:

- if wage above subsistence level ($y > c^*$), then the population of farmers increase $L \uparrow$
- if wage below subsistence level ($y < c^*$), then the population of farmers decrease $L \downarrow$

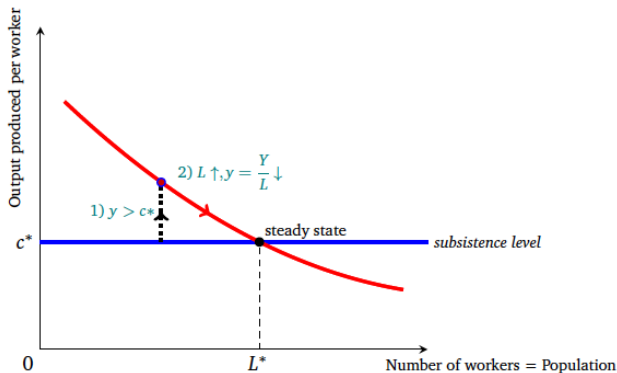
Notice that y is both the living standard (average consumption) and the output per farmer (average production) $\Rightarrow$ think different interpretations of GDP.

The Malthusian model: Dynamics in y



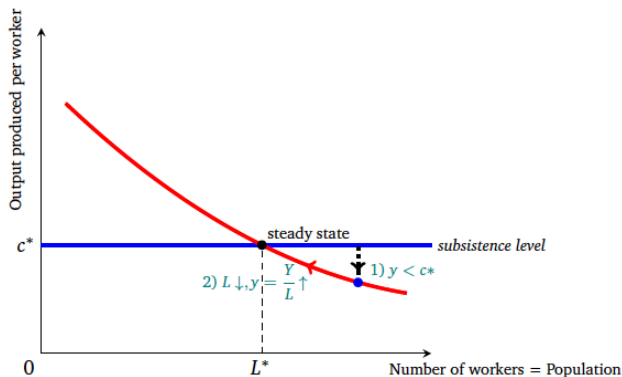
- c^* and L^* represent a **steady state**: the economy has no desire to move from there.
- but if $y > c^*$ due to wage increase, birth rate increases, so L increases (fact 2) + if L increases, then output per farmer $y = Y/L$ has to fall (fact 1), so over time $y \rightarrow c^*$
- also, if $y < c^*$ due to wage decrease, death rate increases, so L decreases (fact 2) + if L decreases, then output per farmer $y = Y/L$ increases (fact 1), so over time $y \rightarrow c^*$

The Malthusian model: Dynamics in y



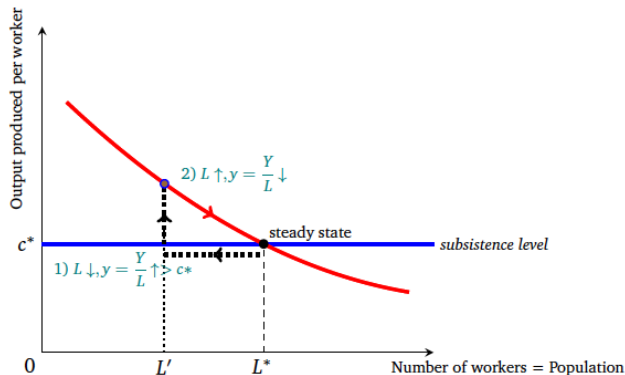
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The Malthusian model: Dynamics in y



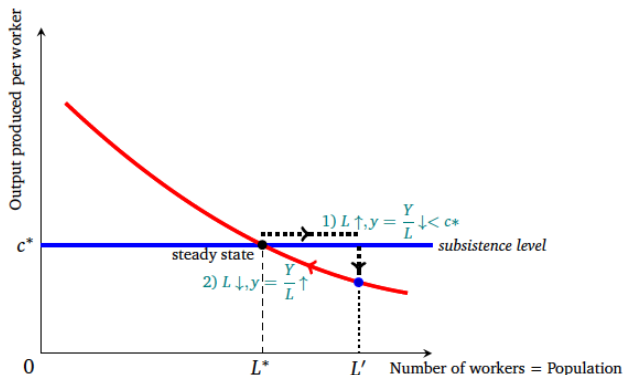
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The Malthusian model: Dynamics in L



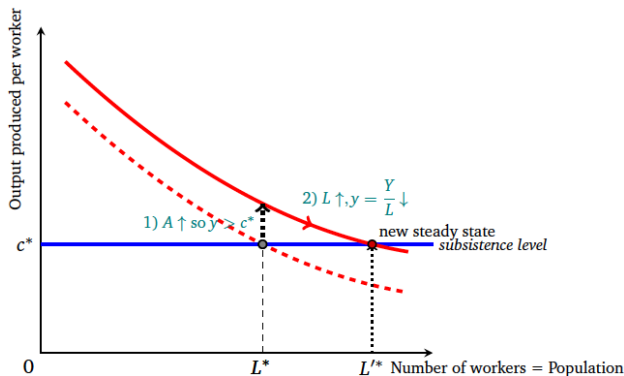
- here, if L were to suddenly fall below L^* , then for fixed land, each farmer has more land = we can get a higher wage (y increases) (fact 1), but this drives up the birth rate, so $L \rightarrow L^*$ (fact 2).
- also, if L were to suddenly rise above L^* , then for fixed land, each farmer has less land = we can get a lower wage (y decreases) (fact 1), this drives up the death rate, so $L \rightarrow L^*$ (fact 2).

The Malthusian model: Dynamics in L



- here, if L were to suddenly fall below L^* , then for fixed land, each farmer has more land = we can get a higher wage (y increases) (fact 1), but this drives up the birth rate, so $L \rightarrow L^*$ (fact 2).
- also, if L were to suddenly rise above L^* , then for fixed land, each farmer has less land = we can get a lower wage (y decreases) (fact 1), this drives up the death rate, so $L \rightarrow L^*$ (fact 2).

The Malthusian model: Dynamics in A



- recall $Y = Af(L)$, so as A increases, so does Y and thus y (the same number of farmers produce more).
- but as y increases (above c^*), L also increases (fact 2).
- and as L increases, then wages have to fall (y decreases), since more farmers are decreasingly productive.
- So y falls until c^* while population has increased: any benefit from the better technology is diluted by the increase in population.

The Solow model: New assumptions

- now: K is no longer constant and can be accumulated: $Y = AF(K, L)$
 - capital is accumulated by **investment**, which is used to replace **depreciated capital**.
 - capital depreciates at a rate $0 < \delta < 1$ per period: if $\delta = 0.2$, then 20% of machines cannot be used after this period.
- now: **constant returns to scale**, i.e. $mF(K, L) = F(mK, mL) \rightarrow$ why? because mathematically expedient.
 - recall CRS means that you can increase EACH factor of production by a factor m and get a production increased by the same factor m .
 - CRS here implies we can rewrite $Y = AF(K, L)$ as $y = Af(k)$, where capital letters are aggregate values and functions and small letters are per-capita values (divided by L) and per-capita functions.

$$\frac{Y}{L} = \frac{AF(K, L)}{L} = A \frac{F(K, L)}{L} \stackrel{\text{CRS}}{=} AF\left(\frac{K}{L}, \frac{L}{L}\right) = AF(k, 1) = Af(k)$$

- unchanged: **diminishing marginal productivity to both capital and labor**, which means $\frac{y}{k}$ is smaller as k is larger.

PS2 Exercise 1. The Solow Model (Part 1)

Consider an economy in which output per worker y depends on the quantity of capital per worker k . The relationship between y and k is given by the following table:

Capital per worker, k	0	1	2	3	4	5
Output per worker, y	0	5	9	12	14	15
Average output per unit of capital						
Marginal output per unit of capital						

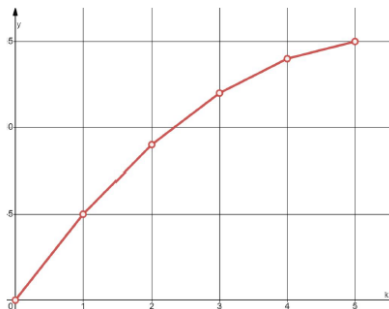
- ① Fill in the table.
- ② Graph y as a function of k .
- ③ Explain how the following variables change as a function of the capital stock per unit of labor:
 - the output per worker
 - average output per unit of capital
 - marginal output per unit of capital

PS2 Exercise 1. The Solow Model (Part 1, solutions)

- 1 Fill in the table.

Capital per worker, k	0	1	2	3	4	5
Output per worker, y	0	5	9	12	14	15
Average output per unit of capital	x	5	4.5	4	3.5	3
Marginal output per unit of capital	x	5	4	3	2	1

- 2 Graph y as a function of k .



PS2 Exercise 1. The Solow Model (Part 1, solutions)

Capital per worker, k	0	1	2	3	4	5
Output per worker, y	0	5	9	12	14	15
Average output per unit of capital	x	5	4.5	4	3.5	3
Marginal output per unit of capital	x	5	4	3	2	1

- 8 Explain how the following variables change as a function of the capital stock per unit of labor:

- the output per worker

output increases in capital: so more machines means higher output.

- average output per unit of capital

average output per unit of capital decreases as capital increases.

- this is a consequence of diminishing marginal productivity of capital.
- since each marginal output is lower than the average output so far, the smaller additional output at each additional unit of capital drags down the average output.
- this is why production is more and more expensive as the quantity increases (marginal cost is increasing).

- marginal output per unit of capital

marginal output per unit of capital is how much each additional unit of capital increases output and is decreasing here → this is the **diminishing marginal productivity of capital**.

- think about how productive a student without a computer would be with a computer; but how much less additional productivity he could have with a second computer.

Solow Model: Math of capital accumulation

Let's hone in on the math here:

- the **capital accumulation** states:

$$k_{t+1} = k_t - \delta k_t + i_t$$

- i.e. capital tomorrow = capital today - what depreciates + what we invest in
- since investment is the saved part of our income and income = output by GDP, we write $i_t = sy_t = sAf(k_t)$ for some fixed savings rate s :

$$k_{t+1} = k_t - \delta k_t + sAf(k_t)$$

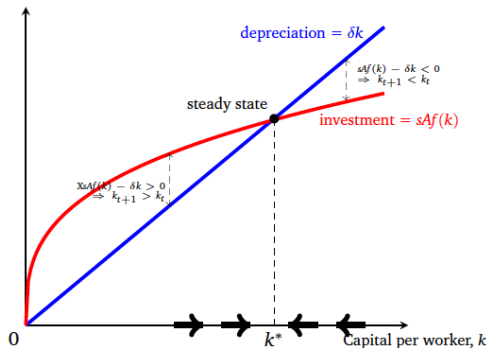
- rearranging, we obtain the law of motion of capital k_t :

$$\underbrace{k_{t+1} - k_t}_{\text{net } \Delta \text{ in capital}} = \underbrace{sAf(k_t) - \delta k_t}_{\text{investment - capital depreciation}}$$

- if $sAf(k_t) > \delta k_t$, then we have invested more than needed to replenish depreciated capital, so capital stock increases in the next period (thus output in the next period increases)
- if $sAf(k_t) < \delta k_t$, then we have invested less than needed to replenish depreciated capital, so capital stock falls in the next period (thus output in the next period decreases)
- if $sAf(k_t) = \delta k_t$, then we have invested just enough to replenish depreciated capital, so capital stock remains unchanged for the next period (thus output remains unchanged)

Solow Model: Steady state

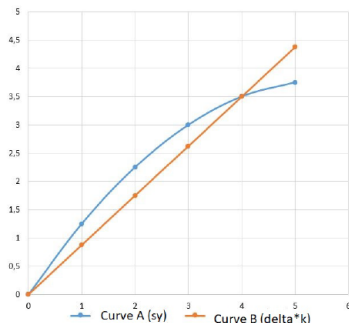
- if $sAf(k_t) > \delta k_t$ for long enough, then output continues to increase.
 - but $f(k_t)$ has diminishing marginal productivity of capital, so any **the investment of new capital** (as a function of the output, i.e. $i_t = sAf(k_t)$) **will also exhibit diminishing marginal productivity of capital**
 $\Rightarrow$ as we accumulate capital, our marginal capital is less and less productive



- at some point, it crosses the depreciation line - each additional invested capital is not enough to offset the depreciation
 $\Rightarrow$ we should stop at the **steady state**.
- in the long run, no capital growth = no output growth

PS2 Exercise 1. The Solow Model (Part 2)

In each period, the capital stock per capita increases because individuals save a part sy of the production. We represent this capital accumulation on the graph below by curve A. On the other hand, a part of the capital δk is destroyed because of obsolescence, represented by curve B.



- ④ Why is the A curve getting flatter and flatter? Explain what the difference between curves A and B tells us about the evolution (positive or negative) of the capital stock per individual.
- ⑤ Towards what stationary level of capital per worker is the economy converging? What happens if the level of capital is below or above this equilibrium level?

PS2 Exercise 1. The Solow Model (Part 2, solutions)

- ④ Why is the A curve getting flatter and flatter? Explain what the difference between curves A and B tells us about the evolution (positive or negative) of the capital stock per individual.

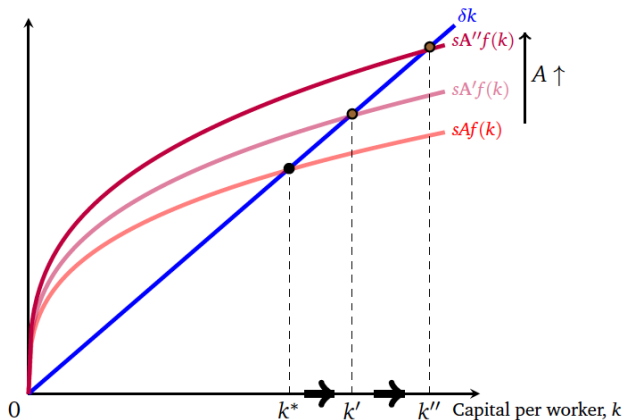
- curve A represents the investment into capital through saved income, sy .
 - y is a function of k and exhibits diminishing MPK as usual: each additional unit of capital is less productive than the previous.
 - and since s is just a constant, it does not affect the diminishing marginal productivity of capital, and thus sy also exhibits diminishing MPK.
- when A is above B, then the investment of capital exceeds the depreciation of capital and the net capital stock per individual increases.
- when A is below B, then the depreciation outstrips the investment of capital and the net capital stock per individual decreases.
- when A meets B at the steady state, then investment is just enough to offset depreciation and so the net capital stock per individual is constant.

- ⑤ Towards what stationary level of capital per worker is the economy converging? What happens if the level of capital is below or above this equilibrium level?

- the steady state is where the curves cross, where $k = k^{SS}$ and constant.
 - if capital is below k^{SS} , then the output generated by the unit of additional investment outweighs that loss by depreciation, so increase capital stock
 - if capital is above k^{SS} , then currently the last unit invested does not bring enough marginal output to offset the depreciation, so we should not have invested that marginal unit and should lower capital stock

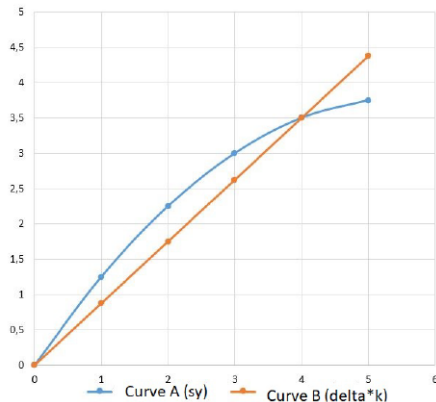
Solow Model: Long-term growth

- We cannot increase k indefinitely along the investment curve due to diminishing MPK.
- How about we shift the curve entirely? How: **increase A** (or increase s ?)



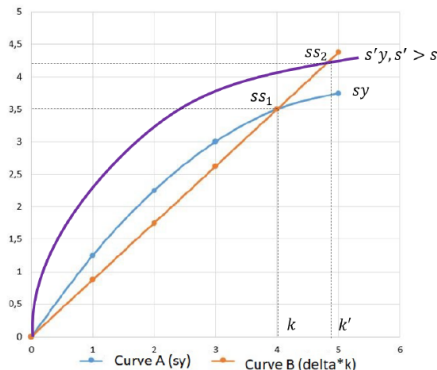
PS2 Exercise 1. The Solow Model (Part 3)

- 6 Show graphically the implications of an increase in the savings rate. How is the stationary level of capital per worker affected?



PS2 Exercise 1. The Solow Model (Part 3, solutions)

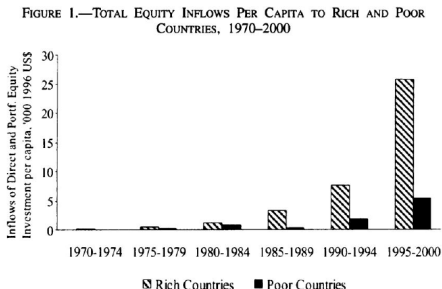
- 6 Show graphically the implications of an increase in the savings rate. How is the stationary level of capital per worker affected?



- higher investment levels shifts the steady state to a higher capital level.

PS2 Exercise 1. The Solow Model (Part 4)

- Consider the graph below describing capital flows according to whether they went to so-called "rich" or "poor" countries between 1970 and 2000. It is taken from an article by Alfaro, Kalemli-Ozcan and Volosovych from 2008.

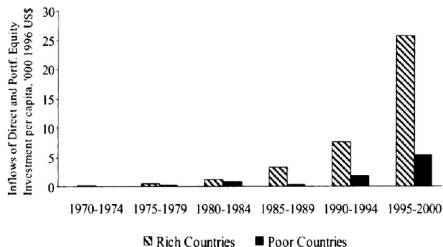


If all countries had the same productivity, would the majority of capital flows go to the so-called "rich" or "poor" countries? What can we deduce from this graph?

PS2 Exercise 1. The Solow Model (Part 4, solutions)

- If all countries had the same productivity, would the majority of capital flows go to the so-called "rich" or "poor" countries? What can we deduce from this graph?

FIGURE 1.—TOTAL EQUITY INFLOWS PER CAPITA TO RICH AND POOR COUNTRIES, 1970–2000



- the proposition: if countries have the same productivities, then we should invest where the MPK is the highest (most bang for our buck) = invest in poor countries
- what do we see: inflows per capita higher in rich countries, and increasingly so
- this is known as the Lucas' puzzle: **if richer countries have lower MPK and poorer countries have higher MPK, in our world of capital flows, we should all investing in poorer countries** - but we don't. Why? → **because our A are different**: the capital itself may have higher returns, but the institutions and the level of existing technology perhaps makes it sub-optimal.

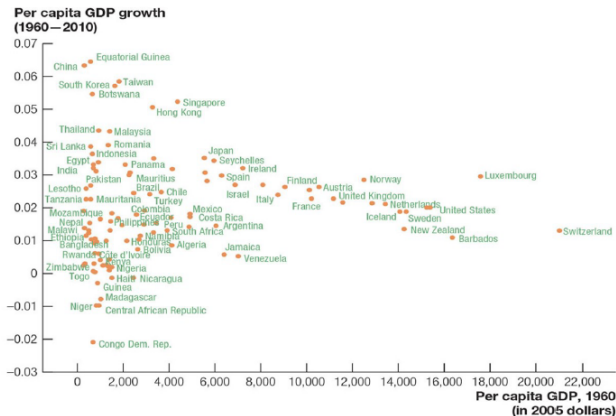
Solow Model: Convergence

- Capital growth rate:

$$g_k = \frac{k_{t+1} - k_t}{k_t} = sA \frac{f(k)}{k} - \delta$$

- So poorer countries have higher MPK and thus higher growth rates, whereas richer countries have lower MPK and thus lower growth rates (since output is driven by capital accumulation)
- Theory: as poorer countries grow, they have lower MPKs and lower growth rates, catching up to the richer countries $\Rightarrow$ (unconditional) convergence
- But this assumes everything else is constant: $s, A, f(k)$ needs to be identical for them to share the same steady state
 $\Rightarrow$ but these parameters aren't usually identical across countries $\Rightarrow$ **conditional convergence**
 - If countries are *institutionally* similar, then we expect convergence $\rightarrow$ they end up at similar steady states
 - Across countries that are not similar, there is no reason to expect convergence $\rightarrow$ they end up at different steady states
- differences in GDP/capita across countries explained by 3 scenarios:
 - similar A , different k (compare France and US, for example)
 - similar k , different A (compare Singapore and Indonesia, for example)
 - different k and different A (compare Germany and Congo, for example)

Solow Model: Convergence



Long-term growth: Institutions and technology

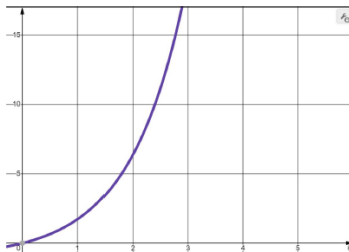
- **institutions:** refer to **any structural element of the economic environment that do not enter production functions but affects the presence and productivity of its factors.**
 - in short, good institutions make it "easier" to produce and create value (rule of law, property rights).
 - bad institutions hinder value creation either by extracting factors/value from the economy (corruption) or by making it difficult to assemble factors well over extended periods of time (poor working conditions, lack of trust, economic and political instability).
- **technology:** refers to **ways of combining factors of production to best generate productive synergies.**
 - early growth can be driven by capital accumulation but Solow model shows that this growth slows down; future growth for high-capital economies can only be driven by **developing technologies or improving institutions.**
 - research drives technological progress (even in the social sciences!) → allow for constant growth, but is costly.
 - so research needs both ideas and protection (**patents**) → but has research become too profit driven (pharmaceuticals?) ← not enough ideas?
 - changing institutions substantially harder: making economic development difficult for developing countries.

PS2 Exercise 2. Production functions

- 1 Draw a graph representing a production function that becomes steeper as the factor of production increases.
- 2 Can you think of an example of a production process that might have this shape? Why would the slope become steeper?
- 3 In this case, what can you say about the marginal and average productivity?

PS2 Exercise 2. Production functions

- ④ Draw a graph representing a production function that becomes steeper as the factor of production increases.



- ② Can you think of an example of a production process that might have this shape? Why would the slope become steeper?

Language learning/skills acquisition. Once you learn the basics, you are able to not only produce more but learn newer skills more easily.

- ③ In this case, what can you say about the marginal and average productivity?

Here, the marginal productivity should increase with each additional unit, which means that the average productivity should also increase (each new unit adds more than the previous unit, so brings up the average).

Next...

- Next seminar is on February 28th.
- **PS3 Exercise 2** due in class.
- **Quiz 2** for next class!
- Take a look at the reading in PS2 and I will upload the answers to the questions on Moodle.